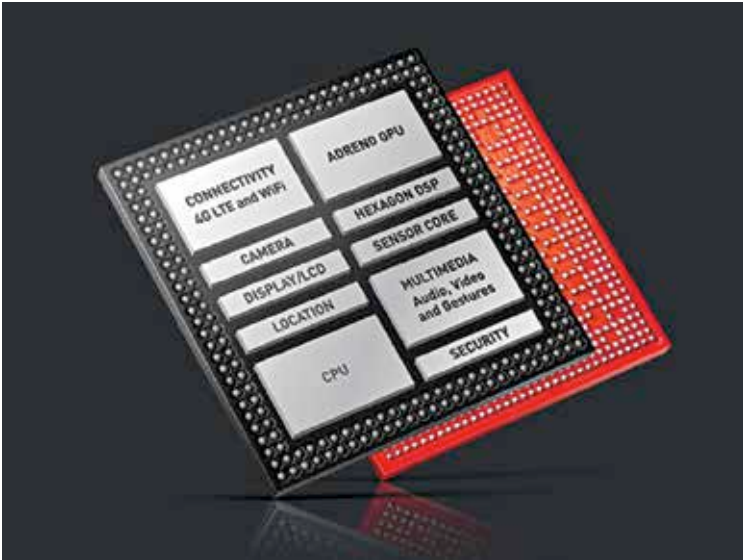




A processor chip from Qualcomm

tially integrate the complete components of a previous end-product and assimilate them into a single chip. These reused components form part of an intellectual property, or IP.

If an IP remains in electronic form, it is one of either two things – a circuit description in HDL or Hardware Description Language (soft IP), or it remains a hardware chip, which includes a hardware IP core. Soft IPs are more flexible, but lack in optimisation. On the other hand, hardware IPs are less flexible, but more optimised.

An SoC usually comprises embedded processors, reused IPs, memory elements like SRAM and ROM, bus architecture, programmable blocks, mixed signal blocks and so on. SoCs may also include other complex components for improved functionality.

However, since IPs belong to other companies, SoC developers have to purchase them from legitimate vendors before incorporating components into the chip. Therefore, it becomes essential to regulate lawful use of IPs to only those who legally own the rights to use them. Otherwise, the IP developer may face loss of revenue. This is why silicon companies have incorporated IP protection measures in VLSI design flow. Security measures have also been applied to SoC design methodology.